

Fundamentals of Security Technology

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Architectural Differences and Security Implications of MBR and GPT

Introduction

The partitioning scheme of a disk plays a crucial role in managing computer storage effectively. Two commonly used partitioning schemes are the Master Boot Record (MBR) and the GUID Partition Table (GPT). In this report, we will explore the architectural differences and similarities between MBR and GPT. We will also investigate the security implications associated with using MBR and GPT in modern operating systems [1].

	MBR	GPT
Maximum capacity	2TB	9.7ZB (~9.7 billion terabytes)
Maximum partitions	26	128
Partition/boot data location	At the beginning of the drive	Throughout the drive
BIOS type	Legacy BIOS	UEFI

Architectural Differences and Similarities between MBR and GPT

MBR and GPT have distinct architectural features that differentiate them from each other [2]. Let's examine the key architectural aspects of MBR and GPT:

Number of Partitions:

MBR: The MBR partitioning scheme supports a maximum of four primary partitions or three primary partitions and one extended partition. Each primary partition is identified by a unique partition table entry in the MBR.

```
DISKPART> list disk

Disk ###  Status       Size      Free      Dyn  Gpt
-----  -
Disk 0    Online      238 GB    3693 MB
Disk 1    Online      57 GB     57 GB

DISKPART> select disk 1

Disk 1 is now the selected disk.

DISKPART> convert mbr

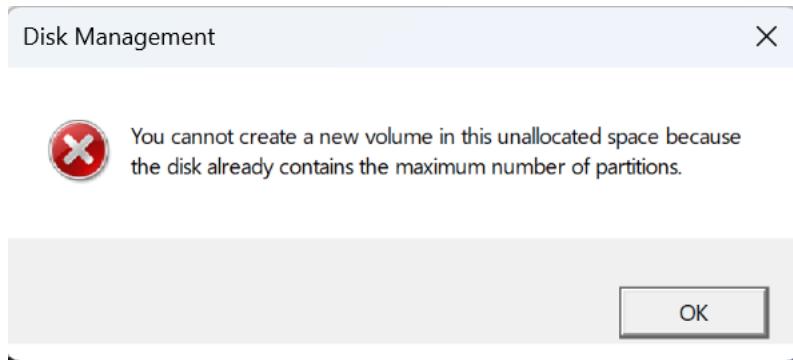
DiskPart successfully converted the selected disk to MBR format.

DISKPART> create partition primary

DiskPart succeeded in creating the specified partition.
```

Disk 1 Removable 57.71 GB Online	(D:) 52.35 GB RAW Healthy (Primary Partition)	Syeda (E:) 1.79 GB NTFS Healthy (Primary Partition)	ROHMA (F:) 1.79 GB FAT32 Healthy (Primary Partition)	SALEEM (G:) 1.79 GB FAT Healthy (Logical Drive)
--	--	--	---	--

Error when tried to create more than 4 volumes:



GPT: The GPT partitioning scheme allows for up to 128 partitions. It utilizes a partition array stored in the GPT header to define the partitions. Each partition has a unique partition GUID that distinguishes it from others.

```
DISKPART> select disk 1

Disk 1 is now the selected disk.

DISKPART> clean

DiskPart succeeded in cleaning the disk.

DISKPART> convert gpt

DiskPart successfully converted the selected disk to GPT format.

DISKPART> list partition
```

Partition ###	Type	Size	Offset
Partition 1	Primary	1829 MB	1024 KB
Partition 2	Primary	1829 MB	1830 MB
Partition 3	Primary	1829 MB	3659 MB
Partition 4	Primary	1829 MB	5488 MB
Partition 5	Primary	1829 MB	7317 MB
Partition 6	Primary	1829 MB	8 GB

Disk 0 Basic 238.46 GB Online								
	529 MB Healthy (Recovery Partition)	100 MB Healthy (EFI System Par	(C:) 233.60 GB NTFS Healthy (Boot, Page File, Crash Dump, Basic Data Partition)		3.60 GB Unallocated	653 MB Healthy (Recovery Partition)		
Disk 1 Removable 57.71 GB Online								
	SYEDA (D:) 1.79 GB FAT Healthy (Basic Data Par	Rohma (E:) 1.79 GB NTFS Healthy (Basic Data Par	SALEEM (F:) 1.79 GB FAT Healthy (Basic Data Par	NEW VOLUME (G:) 1.79 GB FAT Healthy (Basic Data Par	New Volume (H:) 1.79 GB NTFS Healthy (Basic Data Par	NEW VOLUME (I:) 1.79 GB FAT Healthy (Basic Data Par	46.99 GB Unallocated	

Partition Order Management:

MBR: Partition order in MBR is based on the order of entries in the MBR's partition table. The boot loader uses the partition that is currently active in order to start the computer.

In order to keep track of the partitioning, GPT refers to the array of partitions saved in the GPT header.

Altering the order of partitions requires modifying the properties of the partition entry, such as the Partition Priority Index.

Partition Size Information:

MBR: The MBR partitioning technique stores partition sizes in a 32-bit field. This puts a limit on partition sizes of 2 TB.

GPT: To accommodate much greater partition sizes, GPT uses a 64-bit field to record the size of each partition. About 9.4 ZB is the largest partition size that GPT can handle.

Sector Size Information:

MBR: Sector sizes on MBR disks are always 512 bytes. This means that each partition is aligned to 512-byte sectors, and the logical block addressing (LBA) scheme uses 512-byte units.

GPT: GPT supports multiple sector sizes, including 512 bytes, 1024 bytes, and 4096 bytes. The ability to accommodate larger sector sizes improves performance and optimizes disk space utilization.

Volume Name Information:

MBR: MBR does not natively store volume name information within its structure. Instead, the file system on each partition is responsible for maintaining the volume name.

GPT: GPT includes a Unicode name field for each partition, allowing descriptive volume names to be assigned directly to the partitions. This simplifies the identification and management of partitions.

File System Information:

MBR: MBR does not have built-in support for identifying the file system used on a partition. Determining the file system type on an MBR-partitioned disk often requires inspecting the data structures within the file system itself.

GPT: GPT includes a partition type GUID that can be used to identify the file system or the purpose of each partition. This allows for easier identification of the file system without relying solely on the data structures within the file system.

Endian Representation:

MBR: MBR uses little-endian representation for most of its data structures. This means that the least significant byte is stored at the lowest address.

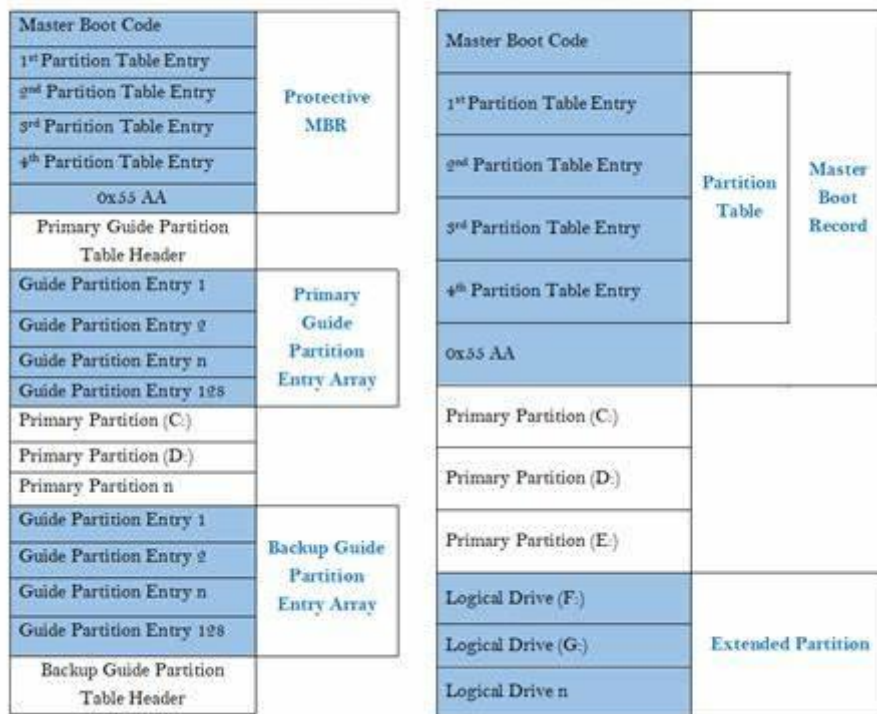
GPT: GPT also uses little-endian representation for most of its data structures, ensuring compatibility with the majority of modern systems.

Data Structures for Booting:

MBR: MBR utilizes a boot sector located in the first sector of the disk. This boot sector contains the boot code and the partition table. The boot code is responsible for locating the active partition and initiating the boot process.

GPT: GPT employs a protective MBR (PMBR) in the first sector to maintain compatibility with legacy systems. The actual boot process in GPT is handled by the EFI System Partition (ESP) and the Extensible Firmware Interface (EFI) boot loaders. The EFI boot loaders are responsible for loading the operating system [3].

GPT VS. MBR Structure



Practical Methodology and Preparation

To gain a practical understanding of MBR and GPT, the following steps were performed [4]:

Configuration:

First Name= Syeda

Middle Name = Rohma

Last Name = Saleem

The size of the partitions will be (Student ID) 19187987* 100 =1918798700 Bytes

We must convert these bytes into megabyte because at the time of partition operating system demand input value in megabyte. so,

$1918798700 \text{ Bytes} / 1024 = 1873826.86 \text{ Kilobytes}$

$1873826.86 \text{ Kilobytes} / 1024 = 1829.90904 \text{ Megabytes}$

So, both drive sizes will be 1829 Megabytes.

$1829.90904 \text{ Megabytes} / 1024 = 1.78702055 \text{ GigaBytes}$

The 1.78 Gigabytes size will be shown on the user interface.

Hardware Setup:

The following hardware configuration was used to analyze disk partitions:

- Computer System: A desktop with adequate processing speed, memory, and storage.
- Hard Drive: The target hard drive that houses the analysis's disk image file.
- External Storage: Backup copies of the disk image and the findings of the analysis might be kept on additional storage devices.

Software Setup:

The following tools and programs have to be installed and set up as part of the software setup:

- Linux Operating System: The analysis's main operating system was installed as Ubuntu 20.04 LTS. It offered a stable and dependable platform with thorough analysis and partitioning functionality for disks.
- Hex Editor: HxD, a well-known hex editor program, was set up on the Linux machine. HxD made it possible to examine and modify the disk image in hexadecimal format.

- **Command-line Tools:** To conduct partitioning operations and acquire data about the disk and its partitions, Linux command-line utilities like fdisk, gdisk, and parted were installed.
- **Virtualization software:** A virtual machine was put up using programs like VirtualBox or VMware in order to establish a secure analytical environment. By separating the analytic environment from the host system, this eliminated the potential for unwanted changes or lost information.

Disk Image Preparation:

Before doing the analysis, the disk image was created using the following methods:

- **Disk Image Creation:** To ensure a precise copy of the original disk, an accurate disk imaging program was utilized to create a disk image of the destination hard drive.
- **Disk Image Integrity Checking** We checked the disk image's integrity using a checksum utility like md5sum or sha256sum. This will allow the imaging to finish without corrupting the disk image.
- **Storage of Disk Image:** The disk image was stored on a separate device from the analysis machine to prevent any unauthorized modifications or corruption.
- **Backup Copy:** A duplicate of the original disk image was made as a backup to guarantee the safety of the data and make it simpler to roll back any accidental changes made during the study.

Through the establishment of a reliable and secure hardware and software environment and the preparation of the disk image, Linux and HxD were able to examine the disk partitions.

MBR Partition:

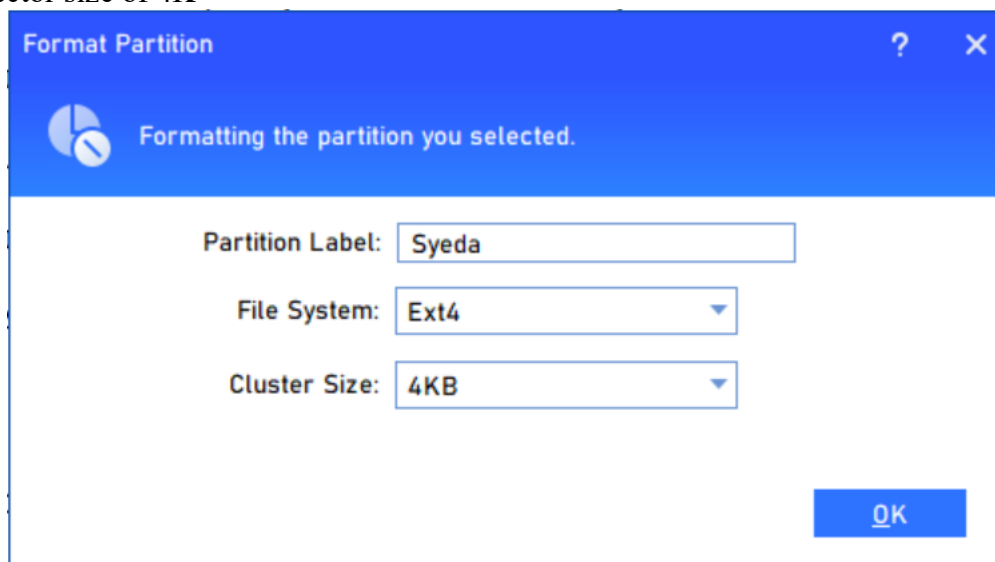
Partitioning hard drives using the Master Boot Record (MBR) technique is a common practice on BIOS-based systems. In 1983, with the release of IBM PC DOS 2.0, MBR became the standard partitioning mechanism. The MBR, the disk's first sector, stores a 32-bit partition table. The Master Boot Record (MBR) stores partition data and the boot loader's instructions [5].

However, there are constraints inherent to the MBR partitioning technique. In terms of main partitions, it can handle up to four, or three with an extended partition. After that, you may split the expanded partition into many logical ones. This restriction, however, becomes problematic when more than four main partitions are required on a disk. The maximum disk capacity is 2 TB since MBR only supports 32-bit logical block addresses.

First Partition:

Format the first partition with Ext4

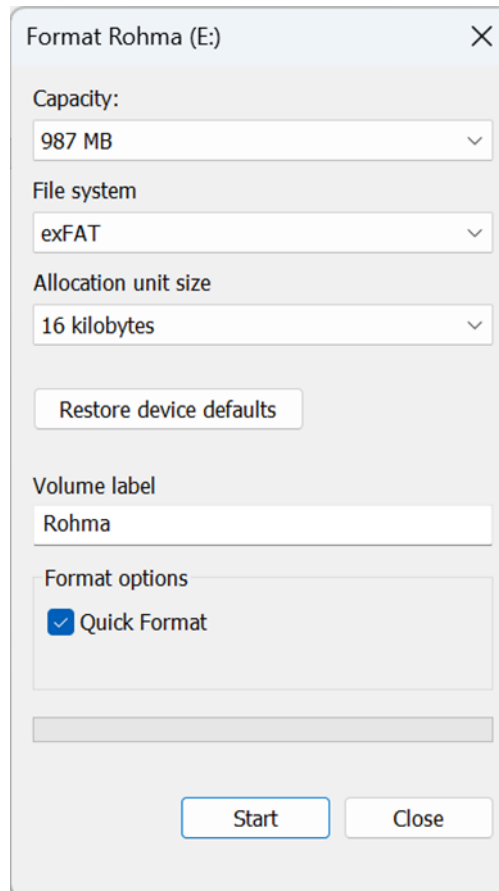
- o Name the first partition your first name
- o Use sector size of 4K



Second Partition:

Format the second partition with exFAT

- o Name the second partition your last name
- o Use sector size of 16K



Format Rohma (E:) X

Capacity:
987 MB

File system
exFAT

Allocation unit size
16 kilobytes

Restore device defaults

Volume label
Rohma

Format options
☒ Quick Format

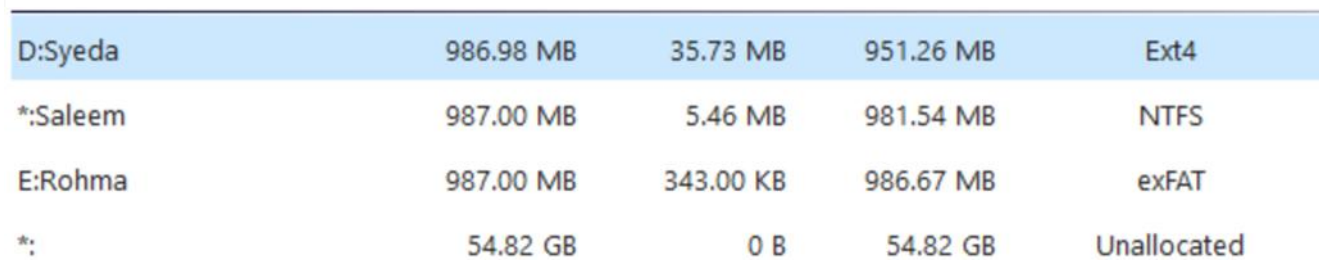
Start Close

Third Partition:

Format the third partition with NTFS

- o Name the second partition your last name
- o Use sector size of 128K

Partition	Format	Partition Name	Sector Size
First	Ext4	Syeda	4K
Second	exFAT	Rohma	16K
Third	NTFS	Saleem	128K



Ext4 (Fourth Extended File System)	exFAT (Extended File Allocation Table)	NTFS (New Technology File System)
- Ext4 is the default file system for most Linux distributions. - It is an extension of the earlier Ext3 file system and provides significant improvements in	- exFAT is a file system developed by Microsoft and is designed to be used on flash drives and external storage devices. - It offers better	- NTFS is the default file system for modern Windows operating systems, including Windows 10. - It is a high-performance

performance and reliability. • Features of Ext4 include support for larger file sizes and partitions, improved journaling for faster file system recovery, and enhanced metadata performance. • Ext4 supports file and directory permissions, symbolic links, and journaling, making it suitable for general-purpose usage in Linux environments. • It is not natively supported by Windows but can be accessed using third-party drivers or software.	compatibility than older FAT file systems and allows for larger file sizes and partitions. • exFAT is supported by both Windows and macOS without the need for additional software or drivers. • It is optimized for use with removable storage devices and supports features such as large file support, file and directory timestamps, and Unicode file names. • exFAT is commonly used for sharing files between Windows and macOS systems or for storing media files on portable storage devices.	file system that provides features such as file and folder permissions, encryption, compression, and journaling for improved data integrity. • NTFS supports large file sizes and partitions and offers advanced features like file and folder-level security, access control lists (ACLs), and transactional updates. • It is not natively supported by macOS but can be accessed using third-party drivers or software. • NTFS is well-suited for use in Windows environments, especially when security and advanced file system features are required.
---	--	--

Linux Commands:

```
sudo fdisk -l
```

```
[parrot@parrot]~$ sudo fdisk -l
Disk /dev/sda: 30 GiB, 32212254720 bytes, 62914560 sectors
Disk model: VMware Virtual S
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xf8d2a63d

Device      Boot Start      End  Sectors  Size Id Type
/dev/sda1   *      2048 62910539 62908492   30G 83 Linux
```

Using this command, you may see the partition maps of all the disks in your system. Type "sudo fdisk -l" to see details about the disk partitions, including their sizes, file systems, and partition types. If you need to execute a command with elevated permissions, you may use the sudo command to do so.

```
sudo dd if=/dev/sda of=mbr.dd bs=512 count=1
```

```
[parrot@parrot]~$ sudo dd if=/dev/sda1 of=mbr.dd bs=512 count=1
1+0 records in
1+0 records out
512 bytes copied, 0.000172428 s, 3.0 MB/s
```

This command use the dd software to make a copy of the /dev/sda disk's Master Boot Record (MBR). /dev/sda, which represents the full disk, is the input file or device specified by the if flag. The of option designates the destination filename for the MBR backup, in this case mbr.dd. Setting the block size to 512 bytes with the bs parameter and specifying 1 for the number of blocks to copy with the count flag (representing the MBR).

```
sudo dd if=/dev/sda of=mbr.dd bs=4096 count=1
```

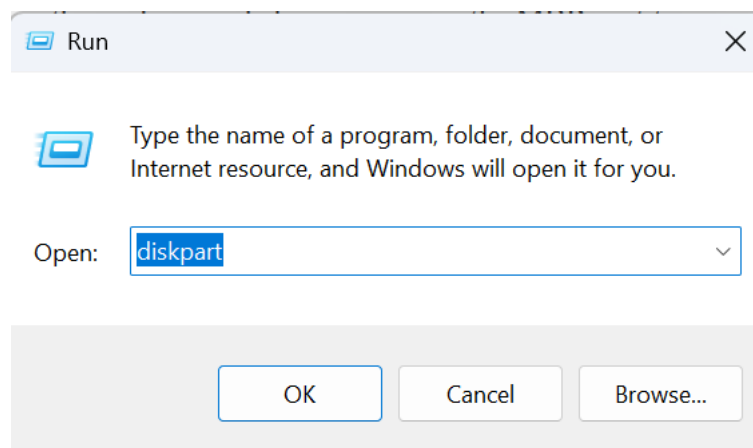
```
[parrot@parrot]~$ sudo dd if=/dev/sda of=mbr.dd bs=4096 count=1
1+0 records in
1+0 records out
4096 bytes (4.1 kB, 4.0 KiB) copied, 0.000450607 s, 9.1 MB/s
```

The difference between this operation and its predecessor is that it increases the block size to 4096 bytes from 512. Data read or written in each operation is defined by its block size. When copying huge files or drives, increasing the block size helps speed up the operation.

```
[parrot@parrot]~$ sudo fdisk -l
Disk /dev/sda: 30 GiB, 32212254720 bytes, 62914560 sectors
Disk model: VMware Virtual S
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xf8d2a63d

Device      Boot Start      End  Sectors  Size Id Type
/dev/sda1   *    2048 62910539 62908492   30G 83 Linux
[parrot@parrot]~$ sudo dd if=/dev/sda1 of=mbr.dd bs=512 count=1
1+0 records in
1+0 records out
512 bytes copied, 0.000172428 s, 3.0 MB/s
[parrot@parrot]~$ sudo dd if=/dev/sda of=mbr.dd bs=4096 count=1
1+0 records in
1+0 records out
4096 bytes (4.1 kB, 4.0 KiB) copied, 0.000450607 s, 9.1 MB/s
[parrot@parrot]~$
```

Then I opened Run window to search for diskpart.



Diskpart is a Windows command-line tool for working with disks, partitions, and volumes. Create, remove, resize, format, and alter disk partitions using its robust range of operations [6]. Diskpart is an invaluable tool for automating and performing complex disk maintenance chores.

Here are some of the most frequent uses of Diskpart that I have encountered:

1. **Disk and Volume Listing:** Diskpart provides a listing of all the disks and volumes on your system. The "list disk" command shows all the disks and their information. The "list volume" command does the same thing for the volumes already present on the drives.
2. **Disk Selection:** Using the "select disk" command and the disk number or identification, we may choose a particular disk on which to do an operation.

3. Partition Management: Diskpart lets you manage partitions on a disk by making new ones, erasing old ones, and modifying their sizes. To make a main partition, use the "create partition primary" command, whereas "create partition extended" does the same thing but with an extended partition. To remove a partition, use the "delete partition" command, whereas "extend" and "shrink" are used to change the size of a partition.
4. Formatting Partitions: Diskpart allows you to use a variety of file systems when formatting partitions. If you want to rapidly set up a partition using the NTFS file system, you may use the "format fs=ntfs quick" command. Other file systems, such as FAT32, exFAT, etc., are interchangeable with "ntfs" [7].

Removing MBR partitioning:

These are the procedures you need to do in Windows to get rid of the MBR partition on a USB drive:

1. If you're using Linux, take out the USB drive.
2. Join the flash drive to your PC's USB port.
3. To bring up the Run box, do the following: Simply use the Windows key plus R to bring up the Run box.
4. Press Enter after typing "diskpart": Using this method, you may access the disk and partition management command-line tool known as DiskPart.
5. To proceed, click "Yes" or provide your administrator credentials if required to do so.
6. List the disks: Just hit Enter after typing "list disk" at the DiskPart prompt. Doing so will show you what storage options you have.
7. Select the USB drive: In the input box, replace "select disk X" with the disk number that corresponds to your USB device, then hit Enter.
8. Clean the disk: Input "clean" and hit the Enter key. Using this command will completely wipe the USB disk of all data and partitions.
9. Convert to GPT: Input "convert gpt" and hit the Enter key. Using this command will change the disk's partitioning strategy from MBR to GPT.
10. Exit DiskPart: To close the DiskPart tool, type "exit" and hit Enter.

Following these instructions will result in the deletion of the MBR partition and the subsequent conversion of the disk to GPT.

```
DISKPART> list disk

Disk ###  Status       Size      Free      Dyn  Gpt
-----  -
Disk 0    Online      238 GB    5522 MB
Disk 1    Online       57 GB     52 GB

DISKPART> select disk 1

Disk 1 is now the selected disk.
```

```
DISKPART> clean

DiskPart succeeded in cleaning the disk.

DISKPART> convert gpt

DiskPart successfully converted the selected disk to GPT format.
```

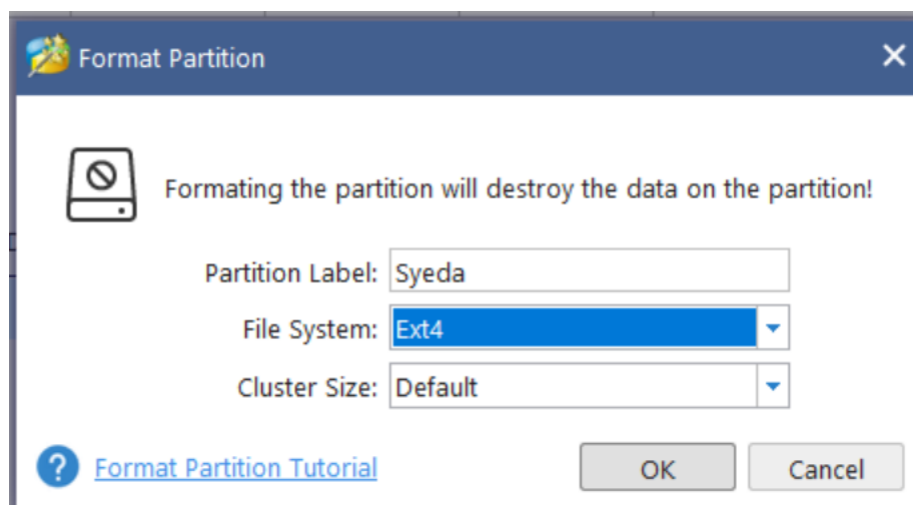
GPT Partition:

If you need to divide up your hard disk, go no further than the cutting-edge GUID Partition Table (GPT) partitioning strategy. It is a component of the UEFI standard, which stands for the Unified Extensible Firmware Interface. GPT is superior to the MBR partitioning technique because it has more useful features and is more adaptable.

GPT assigns a GUID to each disk partition for easy and secure access. Both the partition table in the disk's header and a copy saved in the disk's tail provide this information. By default, GPT allows for up to 128 partitions, making the previous restriction of 4 main partitions obsolete. This facilitates the use of more sophisticated partitioning algorithms and simplifies the administration of huge drives.

Support for disk capacities more than 2 terabytes (TB) is one of GPT's main benefits. GPT makes use of 64-bit logical block addresses, which paves the way for enormous storage capabilities. In addition to providing redundancy, this feature reduces the likelihood of partition table corruption by keeping a copy of the partition table at the disk's end.

The GPT system also includes an MBR for added defense. This ensures compatibility with older systems that use MBR and helps prevent accidental overwriting of GPT partitions by MBR-based tools.



Format Rohma (E:)

Capacity:

987 MB

File system

exFAT

Allocation unit size

16 kilobytes

Restore device defaults

Volume label

Rohma

Format options

☒ Quick Format

StartClose

Format Saleem (F:)

Capacity:

987 MB

File system

NTFS

Allocation unit size

128 kilobytes

Restore device defaults

Volume label

Saleem

Format options

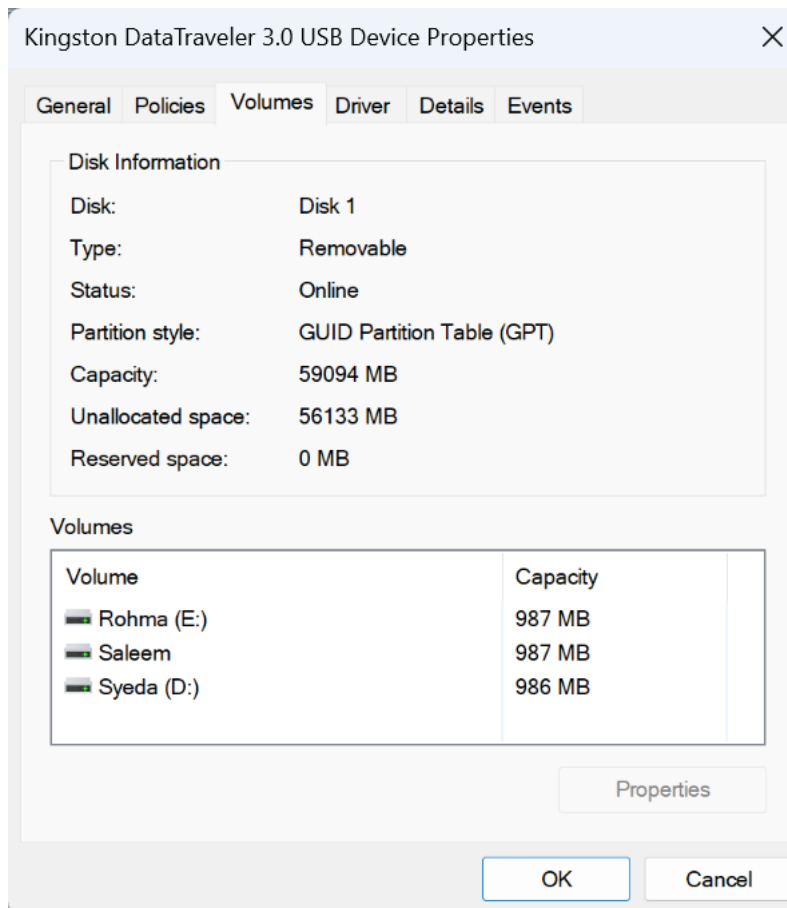
☒ Quick Format

StartClose

Disk 2 (Kingston DataTraveler 3.0 USB, Removable, GPT, 57.71 GB)						
D:Syeda	986.98 MB	34.94 MB	952.05 MB	Ext4	☒ GPT (Data Partition)	None
*:Saleem	987.00 MB	5.46 MB	981.54 MB	NTFS	☒ GPT (Data Partition)	None
E:Rohma	987.00 MB	343.00 KB	986.67 MB	exFAT	☒ GPT (Data Partition)	None
*:	54.82 GB	0 B	54.82 GB	Unallocated	☐ GPT	None

Disk 1	Recovery(N	(FAT32)	(Other)	C:(NTFS)	(Unallocate	(NTFS)
GPT 238.47 GB	529 MB (Usi	100 MB (Usi	16 MB	231.8 GB (Used: 51%)	5.4 GB	653 MB (Usi

Disk 2	D:Syeda(Ext	Saleem(NTF	E:Rohma(ex	(Unallocated)
GPT 57.71 GB	987 MB (Usi	987 MB (Usi	987 MB (Usi	54.8 GB



GPT Disk Creation:

```
[parrot@parrot]~[-]
$ sudo dd if=/dev/sda1 of=gpt_image.dd bs=512 count=34 skip=1
[sudo] password for parrot:
34+0 records in
34+0 records out
17408 bytes (17 kB, 17 KiB) copied, 0.00049051 s, 35.5 MB/s
[parrot@parrot]~[-]
$ sudo dd if=/dev/sda1 of=gpt_image.dd bs=4096 count=34 skip=1
34+0 records in
34+0 records out
139264 bytes (139 kB, 136 KiB) copied, 0.00134533 s, 104 MB/s
[parrot@parrot]~[-]
$
```

HxD:

A well-liked hex editor program for reading and editing binary files is called HxD. Data analysis, file editing, disk imaging, and forensic investigations are just a few of the many tasks it is useful for because it offers such a wide range of capabilities and functions. An outline of HxD and its main characteristics may be found in the following sections.

1. User Interface:

Users of HxD can efficiently browse and modify binary files because to its user-friendly and straightforward interface. The hexadecimal view is on the left, while the matching ASCII representation is on the right, in the main window's two panels. Users may concurrently evaluate the binary data in hex and text representations thanks to the dual-view layout.

2. Hexadecimal and ASCII Views:

HxD shows a file's contents in the hexadecimal view, byte-by-byte displaying the binary data. It is simpler to understand and evaluate text-based data within the file thanks to the accompanying ASCII view, which offers the relevant characters for each byte.

3. Data Analysis and Editing:

Users may carry out a number of actions on binary files using HxD. These include scouring the file for

certain patterns or values, changing data, copying and pasting specific file segments, and adding new data. The program enables effective data manipulation by supporting both automated batch processes and human editing.

4. File Comparison and File Concatenation:

Users of HxD can contrast two binary files side by side to see their differences. When comparing multiple versions of a file or recognizing changes made to a file, this functionality is especially helpful. Additionally, HxD gives users the option to combine numerous files into a single binary file by concatenating them.

5. Disk and Memory Editing:

HxD offers disk and memory editing in addition to file editing. Users can open disk image files and edit and analyze the images directly using this software. Forensic investigations, disk imaging, and recovery jobs all benefit from this functionality.

6. Structure Viewer and Analyzer:

A structure viewer provided by HxD enables users to create unique data structures for intricate file types. This feature makes it possible to analyze and visualize structured data inside of files, giving users insights into the structure and design of particular file formats.

7. Checksum Calculation and File Shredding:

Several checksum techniques are supported by HxD, allowing users to compute checksum values (such as CRC32, MD5, and SHA-1) for data integrity verification. The program also includes a file-shredding function that enables users to securely delete sensitive data by overwriting it with random numbers.

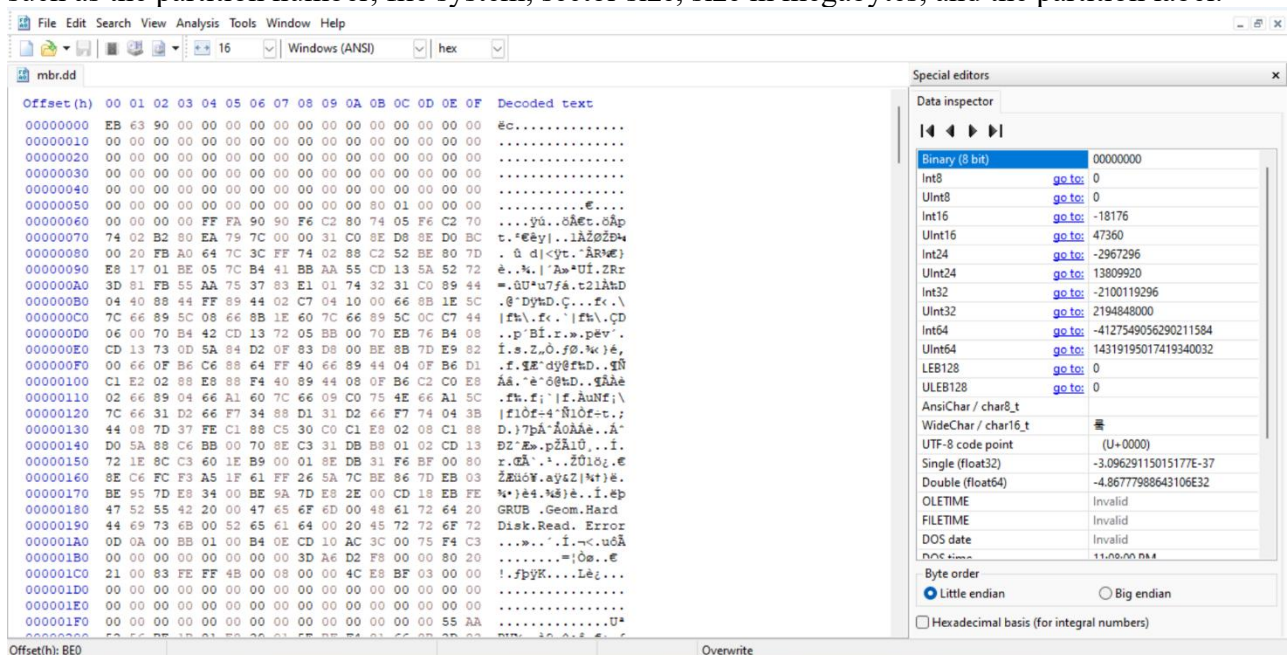
8. Plug-ins and Customization:

HxD accepts plug-ins that increase its capability and let users include unique features and functionalities. Additionally, the software has possibilities for customization, letting users to up the appearance, design, and behavior to suit their tastes.

MBR Table:

Different methods are used to handle partition data via the MBR (Master Boot Record) and GPT (GUID Partition Table). The partition count is determined by bytes 80 to 83 of the GPT. In GPT, partition entries are distinct from the header and take up 128 bytes per partition. The header of the MBR contains the partition entries, each of which takes up 16 bytes and is located between bytes 446 and 509.

In the MBR partition table, each partition is represented by an entry. The table contains information such as the partition number, file system, sector size, size in megabytes, and the partition label.



When I opened the mbr.dd image file in HxD tool, the result of that was the hexadecimal, offset, ASCII values that helps in analyzing the MBR image.

File Edit Search View Analysis Tools Window Help																	
16 Windows (ANSI) hex																	
mbr.dd																	
Offset(h)	00	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F	Decoded text
00000000	EB	63	90	00	00	00	00	00	00	00	00	00	00	00	00	00	ëc.....
00000010	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
00000020	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
00000030	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
00000040	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
00000050	00	00	00	00	00	00	00	00	00	00	00	80	01	00	00	00	ë....
00000060	00	00	00	00	FF	FA	90	90	F6	C2	80	74	05	F6	C2	70	ýú...ôÀæt.ôÀp
00000070	74	02	B2	80	EA	79	7C	00	00	31	C0	8E	D8	8E	D0	BC	t.°ëÿ ...lÀžØžĐ¼
00000080	00	20	FB	A0	64	7C	3C	FF	74	02	88	C2	52	BE	80	7D	. û d <ýt.°ÀR¾ë}
00000090	E8	17	01	BE	05	7C	B4	41	BB	AA	55	CD	13	5A	52	72	è..¾. °A»°Uí.ZRr
000000A0	3D	81	FB	55	AA	75	37	83	E1	01	74	32	31	C0	89	44	=.ûU°u7fá.t2lÀ¾D
000000B0	04	40	88	44	FF	89	44	02	C7	04	10	00	66	8B	1E	5C	.@°Dý%D.Ç...f<.\
000000C0	7C	66	89	5C	08	66	8B	1E	60	7C	66	89	5C	0C	C7	44	f%\.f<.` f%\.ÇD
000000D0	06	00	70	B4	42	CD	13	72	05	BB	00	70	EB	76	B4	08	..p°BÍ.r.».pëv´.
000000E0	CD	13	73	0D	5A	84	D2	0F	83	D8	00	BE	8B	7D	E9	82	Í.s.Z„Ö.fØ.¾}é.
000000F0	00	66	0F	B6	C6	88	64	FF	40	66	89	44	04	0F	B6	D1	.f.¶E°dý@f%D..¶Ñ
00000100	C1	E2	02	88	E8	88	F4	40	89	44	08	0F	B6	C2	C0	E8	ÁÁ.°è°ô@%D..¶ÀÀè
00000110	02	66	89	04	66	A1	60	7C	66	09	C0	75	4E	66	A1	5C	.f%.f;` f.ÀuNf;\ f1òf÷4°Ñlòf÷t.;
00000120	7C	66	31	D2	66	F7	34	88	D1	31	D2	66	F7	74	04	3B	D.)7pÁ°ÀOÀÀè..Á°
00000130	44	08	7D	37	FE	C1	88	C5	30	C0	C1	E8	02	08	C1	88	ĐZ°E».pžĂlŮ...í.
00000140	D0	5A	88	C6	BB	00	70	8E	C3	31	DB	B8	01	02	CD	13	r.ĠĂ°.°...žŮlôç.ë
00000150	72	1E	8C	C3	60	1E	B9	00	01	8E	DB	31	F6	BF	00	80	žÆüó¥.ay&Z ¾†}ë.
00000160	8E	C6	FC	F3	A5	1F	61	FF	26	5A	7C	BE	86	7D	EB	03	žÆüó¥.ay&Z ¾†}ë.
00000170	BE	95	7D	E8	34	00	BE	9A	7D	E8	2E	00	CD	18	EB	FE	¾*}è4.¾š}è..í.ëp
00000180	47	52	55	42	20	00	47	65	6F	6D	00	48	61	72	64	20	GRUB .Geom.Hard
00000190	44	69	73	6B	00	52	65	61	64	00	20	45	72	72	6F	72	Disk.Read. Error
000001A0	0D	0A	00	BB	01	00	B4	0E	CD	10	AC	3C	00	75	F4	C3	...»...´.í.¬<.uôĂ
000001B0	00	00	00	00	00	00	00	00	3D	A6	D2	F8	00	00	80	20	= òø..ë
000001C0	21	00	83	FE	FF	4B	00	08	00	00	4C	E8	BF	03	00	00	!.fpÿK....Lèç...
000001D0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
000001E0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
000001F0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	55	AA	U°

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16 Windows (ANSI) hex																	
gpt final.dd mbr final.dd																	
Offset(h)	00	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F	Decoded text
00000E00	FF	FF	E2	F6	58	58	E9	E8	FE	FF	FF	8B	45	EC	05	D8	ÿÿäöXXéèþÿÿ<Ei.Ø
00000E10	00	00	00	E8	D5	FD	FF	FF	8B	55	E4	73	1E	8B	45	EC	...èðÿÿÿ<Uàs.<Ei
00000E20	05	E4	00	00	00	E8	C3	FD	FF	FF	8B	55	E0	73	06	8B	.ä...èÃÿÿÿ<Uàs.<
00000E30	55	E0	87	55	DC	FF	75	E4	8F	45	E0	87	55	E8	89	55	Uà+UÜÿuä.Eà+Uè+U
00000E40	E4	B8	34	05	00	00	E8	1A	FE	FF	FF	80	7D	EC	07	C6	ä,4...è.pÿÿÿ}i.Æ
00000E50	45	EC	08	72	04	80	45	EC	03	E9	AE	00	00	00	8B	45	Ei.r.€Ei.éø...<E
00000E60	E8	87	45	E4	87	45	E0	89	45	DC	80	7D	EC	07	C6	45	è+Eä+Eà+EÜÿ}i.ÆE
00000E70	EC	07	72	04	80	45	EC	03	B8	32	03	00	00	E8	E3	FD	i.r.€Ei.,2...èäÿ
00000E80	FF	FF	52	B8	03	00	00	00	39	C2	76	02	89	C2	B1	06	ÿÿR,...9Äv.%Ä±.
00000E90	D3	E2	8D	82	B0	01	00	00	E8	9C	FD	FF	FF	89	55	E8	Öä.,°...èæÿÿÿUè
00000EA0	83	FA	04	72	63	89	D1	D1	E9	49	0F	B6	C2	24	01	0C	fú.rcwÑÑéI.qÄ\$.
00000EB0	02	D3	E0	89	45	E8	83	FA	0E	73	0C	8B	45	E8	05	AF	.Öä+Eèfú.s.<Eè.~
00000EC0	02	00	00	29	D0	EB	39	80	E9	04	31	D2	D1	6D	F4	D1	...)Ðè9€é.lÖÑmÖÑ
00000ED0	E2	8B	45	F4	39	45	F0	72	06	29	45	F0	80	CA	01	3D	ä(Eö9EöR.)Eö€E.=
00000EE0	00	00	00	01	73	0C	C1	65	F4	08	C1	65	F0	08	AC	88	s.Äèö.Äèö.~^
00000EF0	45	F0	E2	D8	B1	04	D3	E2	01	55	E8	B8	22	03	00	00	Eöäö±.Öä.Uè."...
00000F00	E8	34	FD	FE	FF	01	4D	E8	FF	45	E8	5A	83	C2	02	89	è4ÿÿÿ.MèÿEèZfÄ.%
00000F10	D1	E9	DF	FE	FF	FF	8D	B4	26	00	00	00	00	8D	76	00	Ñèäþÿÿ.~ä....v.
00000F20	00	44	A3	83	DF	0C	B3	4D	0D	CF	B8	D6	3F	CD	FE	54	.Dèfä.%M.ï.Ö?ÍþT
00000F30	1C	91	20	D1	A9	62	46	8C	D3	75	F4	05	FC	9E	BD	09	.~ÑèbF€Öuö.üž±.
00000F40	6E	05	6B	CA	25	8D	12	75	4D	23	11	2D	3B	61	9D	96	n.kÈ%.uM#.-;a.-
00000F50	ED	D4	EB	FB	7C	A2	EA	F3	EC	F3	A0	67	F0	FB	96	19	iÖèü cèöiö göü-.~
00000F60	35	D3	D9	08	07	80	57	A3	6A	76	C9	7E	97	9C	10	52	5ÖÜ..€WèjvÈ~æ.R
00000F70	AF	E6	D7	C5	4D	3B	F5	22	5B	BC	D6	56	6E	60	73	3E	~æ*ÄM;ö"[%ÖVn's>
00000F80	4C	CD	98	4A	57	C9	B3	94	CD	A5	CD	2B	AD	B9	7A	0D	Lí~JWÈ%~Íxí+.~z.
00000F90	46	C3	90	2A	41	DD	74	EB	1A	EE	72	50	0C	45	89	3D	FÄ.*AÝtè.irP.E%~
00000FA0	56	BA	5F	A6	4F	43	4C	D3	60	24	2C	A9	B3	EE	96	DC	V°_!OCLÖ's,è~i-Ü
00000FB0	C4	59	A0	02	D9	8C	BA	7A	24	57	C3	66	D9	38	C6	2B	ÄY .Ü€°z\$WÄfÜöE+
00000FC0	D4	FB	F0	5E	7C	D7	10	F6	D1	72	13	A6	14	0A	64	95	Öüö^ *.öÑr. .d*
00000FD0	01	01	5D	17	A0	5A	A9	A8	B4	F1	91	2C	00	62	C1	39	..].Z€~"Ä",.bÄ9
00000FE0	EB	F8	78	DF	9D	84	A4	75	C7	16	D8	3D	85	E4	4A	58	èøxä.,~uÇ.Ø=...äJX
00000FF0	EF	73	FB	A9	08	BD	2A	CB	05	39	00	7D	E3	92	0E	7F	Ñsüö.%*È.9.)ä'..

The number of sectors, which is contained in bytes 12 to 15 of the partition sector, determines the size of a partition in the MBR. The byte range "00 58 39 00" in the MBR DD image denotes the partition's size. This value has to be changed to big-endian because it is currently in little-endian format. The result of conversion is "00 39 58 00" in hexadecimal. This hexadecimal number, "00 39 58 00," has to be translated into decimal in order to determine the size.

GPT Table:

00000200	45	46	49	20	50	41	52	54	00	00	01	00	5C	00	00	00	EFI PART....\...
00000210	97	88	0F	B6	00	00	00	00	01	00	00	00	00	00	00	00	-^.%.....
00000220	FF	BF	E0	01	00	00	00	00	22	00	00	00	00	00	00	00	ÿ¿à.....".....
00000230	DE	BF	E0	01	00	00	00	00	37	A8	80	62	35	6B	8D	40	È¿à.....7"€b5k.€
00000240	97	01	64	9A	87	29	0E	F1	02	00	00	00	00	00	00	00	-.dš±).ñ.....
00000250	80	00	00	00	80	00	00	00	F3	8B	74	D0	00	00	00	00	€ € €...+ft

```

00000400  A2 A0 D0 EB E5 B9 33 44 87 C0 68 B6 B7 26 99 C7  c 3D+Ahq-gm
00000410  FF D8 5C D3 9F B3 3C 48 84 D9 B0 8E 01 42 8D A1  y0\OY'<H„Ü°Z.B.
00000420  00 08 00 00 00 00 00 00 FF 5F 39 00 00 00 00 00  .....y_9.....
00000430  00 00 00 00 00 00 00 00 42 00 61 00 73 00 69 00  .....B.a.s.i.
00000440  63 00 20 00 64 00 61 00 74 00 61 00 20 00 70 00  c. .d.a.t.a. .p.
00000450  61 00 72 00 74 00 69 00 74 00 69 00 6F 00 6E 00  a.r.t.i.t.i.o.n.
00000460  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  .....
00000470  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  .....
00000480  A2 A0 D0 EB E5 B9 33 44 87 C0 68 B6 B7 26 99 C7  c 3D+Ahq-gm
00000490  3E BF E1 64 64 D3 29 48 8A 83 1F EC B5 C0 24 29  >¿áddÓ)HŠf.ìpÀ$)
000004A0  00 60 39 00 00 00 00 00 FF B7 72 00 00 00 00 00  .`9.....y·r.....
000004B0  00 00 00 00 00 00 00 00 42 00 61 00 73 00 69 00  .....B.a.s.i.
000004C0  63 00 20 00 64 00 61 00 74 00 61 00 20 00 70 00  c. .d.a.t.a. .p.
000004D0  61 00 72 00 74 00 69 00 74 00 69 00 6F 00 6E 00  a.r.t.i.t.i.o.n.
000004E0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  .....

```

The quantity of sectors a partition takes up in the MBR often indicates how big it is. The partition size details are represented by the bytes "00 58 39 00" in the given MBR DD image. Because the bytes are organized differently than their real values, this value is saved in little-endian format. We must translate the little-endian representation into big-endian in order to get the right value. The value is converted to "00 39 58 00" in hexadecimal form after the conversion. To ascertain the exact size, let's now convert this hexadecimal value to decimal. The decimal equivalent of "00 39 58 00" is 145,408. This shows that there are 145,408 sectors in the partition. Depending on how the disk's sector size is configured, each sector's size may differ. We can evaluate the allocation of disk space and learn more about the partitioned disk's storage capacity by studying the partition size information in the MBR.

```

000001C0  21 00 0C 0E 24 EA 00 08 00 00 00 58 39 00 00 0E  !...$è....X9...

```


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16 Windows (ANSI) hex																	
gpt final.dd mbr final.dd																	
Offset (h)	00	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F	Decoded text
0000FB20	00	00	00	00	00	00	00	00	00	00	00	00	80	66	0C	00	ef..
0000FB30	00	00	00	75	05	00	00	00	00	00	00	00	C0	54	01	00	...u.....ÀT..
0000FB40	00	00	00	EA	01	00	00	00	00	00	00	00	80	5C	0C	00	...ê.....e\..
0000FB50	00	00	00	75	05	00	00	00	00	00	00	00	40	D5	01	00	...u.....@õ..
0000FB60	00	00	00	0B	00	00	00	00	00	00	00	00	40	53	02	00	@S..
0000FB70	00	00	00	61	05	00	00	00	00	00	00	00	80	5F	0B	00	...a.....€_..
0000FB80	00	00	00	71	05	00	00	00	00	00	00	00	90	D0	7F	07	...q.....Đ..
0000FB90	00	00	00	00	B0	42	09	04	00	00	00	00	01	00	00	00	°B.....
0000FBA0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
0000FBB0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
0000FBC0	00	00	00	01	00	02	00	00	02	00	00	00	00	00	00	00	
0000FBD0	00	00	00	00	80	DD	01	00	00	00	00	76	05	00	00	00	...éý.....v....
0000FBE0	00	00	00	00	C0	54	01	00	00	00	00	EA	01	00	00	00	...ÀT.....ê....
0000FBF0	00	00	00	00	80	77	0E	00	00	00	00	76	05	00	00	00	...€w.....v....
0000FC00	00	00	00	00	40	D5	01	00	00	00	00	0B	00	00	00	00	...@õ.....
0000FC10	00	00	00	00	40	53	02	00	00	00	00	61	05	00	00	00	...@S.....a....
0000FC20	00	00	00	00	40	D1	01	00	00	00	00	76	05	00	00	00	...@Ñ.....v....
0000FC30	00	00	00	00	90	D0	7F	07	00	00	00	00	00	43	09	04	Đ.....C..
0000FC40	00	00	00	01	00	00	00	00	00	00	00	00	00	00	00	00	
0000FC50	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
0000FC60	00	00	00	00	00	00	00	00	00	00	00	00	01	00	02	00	
0000FC70	02	00	00	00	00	00	00	00	00	00	00	00	40	51	0C	00	@Q..
0000FC80	00	00	00	73	05	00	00	00	00	00	00	00	C0	54	01	00	...s.....ÀT..
0000FC90	00	00	00	EA	01	00	00	00	00	00	00	00	C0	4B	0C	00	...ê.....ÀK..
0000FCA0	00	00	00	73	05	00	00	00	00	00	00	00	40	D5	01	00	...s.....@õ..
0000FCB0	00	00	00	0B	00	00	00	00	00	00	00	00	40	53	02	00	@S..
0000FCC0	00	00	00	61	05	00	00	00	00	00	00	00	80	5F	0B	00	...a.....€_..
0000FCD0	00	00	00	71	05	00	00	00	00	00	00	00	90	D0	7F	07	...q.....Đ..
0000FCE0	00	00	00	00	B0	42	09	04	00	00	00	00	01	00	00	00	°B.....
0000FCF0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
0000FD00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
0000FD10	00	00	00	01	00	02	00	00	02	00	00	00	00	00	00	00	

Analysis of Partitioning Information

The screenshots and annotations reveal the similarities and differences in the partitioning information stored in MBR and GPT. The screenshots showcase the partition sizes, file system types, sector sizes, volume names, and other relevant information for each partition. By examining these details, a comprehensive understanding of how MBR and GPT differ architecturally can be achieved.

Attacker Techniques to Complicate MBR Rebuilding

Attackers may use a variety of methods to make a successful MBR rebuild much more difficult, if not impossible. Examples of such methods are:

- **Malware and Rootkits:** Attackers may infect the system with malware or rootkits that attack the MBR. Malicious software has the potential to alter or replace the MBR, making it very difficult to revert to the disk's original layout.
- **Bootkit Attacks:** Bootkits are sophisticated malware that compromises the boot process. They are located in the MBR or boot sector and may modify the boot process, making it impossible to rebuild the MBR without specialist tools or knowledge.
- **Disk-Wiping Attacks:** Attackers may deploy disk-wiping methods that delete the whole disk, including the MBR and partitioning information. When this occurs without adequate

safeguards, the process of rebuilding the MBR becomes impossible.

- **Advanced Persistence Threats (APTs):** APTs are stealthy and persistent threats that aim to maintain a presence on compromised systems over an extended period. These threats may specifically target the MBR, making it challenging to recover the original disk structure.

Conclusion

In conclusion, understanding the architectural differences and similarities between MBR and GPT is crucial for effective disk management and system security. MBR, with its limitations in partition size and number, has been widely used for decades. However, GPT provides significant advancements, including support for larger disk capacities, more partitions, and enhanced data integrity. GPT also offers improved security features, such as protective MBR and support for larger sector sizes.

The practical exercise of creating partitions using MBR and GPT, along with the analysis of the partitioning information, provided a hands-on experience to reinforce the theoretical knowledge. The screenshots and observations highlighted the various aspects of MBR and GPT, including partition sizes, file system types, sector sizes, and volume names.

Furthermore, it is important to consider the security implications associated with using MBR and GPT. While GPT offers enhanced security features, attackers can employ various techniques, such as malware infections and disk-wiping attacks, to complicate the process of rebuilding the MBR.

By studying the architectural differences, security implications, and recovery techniques, we can make informed decisions when choosing between MBR and GPT partitioning schemes for modern operating

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